

$$\exists x \in \mathbb{Z} : \forall y \in \mathbb{Z} : x < y \quad \text{F}$$

$$\neg (\exists x \in \mathbb{Z} : \forall y \in \mathbb{Z} : x < y)$$

$$\equiv \forall x \in \mathbb{Z} : (\forall y \in \mathbb{Z} : x < y)$$

$$\equiv \forall x \in \mathbb{Z} : \exists y \in \mathbb{Z} : x \leq y$$

$$\forall x \in \mathbb{Z} : \exists y \in \mathbb{Z} : \forall z \in \mathbb{Z} : \frac{x}{y} \leq z \quad \text{T}$$

for every $x \in \mathbb{Z}^+$
find a $y \in \mathbb{Z}^+$ such that
for every $z \in \mathbb{Z}^+$:
 $\frac{x}{y} \leq z \quad x < y$

$$\forall x \in \mathbb{Z} : \exists y \in \mathbb{Z} : \forall z \in \mathbb{Z} : \frac{x}{y} \leq z \quad \text{F}$$

for every $z \in \mathbb{Z}^+$
find a $y \in \mathbb{Z}^+$ such that
for every $x \in \mathbb{Z}^+$:
 $\frac{x}{y} \leq z \quad (x > yz)$

$$\frac{y \cdot z + 1}{y} \leq z$$

$$\underline{\underline{z+1 \leq z}}$$

$$n \text{ is odd} \Rightarrow n^2 \text{ is odd}$$

$$((p \Rightarrow q) \wedge (p_1 \Rightarrow p_2) \wedge \dots \wedge (p_n \Rightarrow q)) \Rightarrow (p \Rightarrow q)$$

$$n \text{ odd}$$

$$\Leftrightarrow \exists k \in \mathbb{Z} : n = 2k+1$$

$$\Rightarrow \exists k \in \mathbb{Z} : n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$$

$$\Rightarrow \exists k' \in \mathbb{Z} : n^2 = 2k' + 1$$

$$\Leftrightarrow n^2 \text{ odd}$$

qed

$$n^2 \text{ odd} \Rightarrow n \text{ odd}$$

$$\neg (n \text{ odd})$$

$$n \text{ even}$$

$$\Rightarrow \exists k \in \mathbb{Z} : n = 2k$$

$$\Rightarrow \exists k \in \mathbb{Z} : n^2 = 4k^2 = 2(2k^2)$$

$$\Rightarrow \exists k' \in \mathbb{Z} : n^2 = 2k'$$

$$\Leftrightarrow n^2 \text{ even}$$

$$\Rightarrow \neg (n^2 \text{ odd})$$

$$\neg (n \text{ odd})$$

$$n \text{ even}$$

$$(p \Rightarrow q) \equiv (\neg p \Rightarrow \neg q)$$

$$n^2 \text{ is odd} \Leftrightarrow n \text{ is odd}$$

$\sqrt{2} \notin \mathbb{Q}$
no proof, but classical
proof by contradiction

Let $n \in \mathbb{Z}$ then.

$$\left\lfloor \frac{n+1}{2} \right\rfloor \geq \frac{n}{2}$$

case 1: n odd

$$n \text{ odd} \Rightarrow n+1 \text{ even} \Rightarrow \left\lfloor \frac{n+1}{2} \right\rfloor = \frac{n+1}{2} \geq \frac{n}{2}$$

case 2: n even

$$n \text{ even} = n+1 \text{ odd} \Rightarrow \left\lfloor \frac{n+1}{2} \right\rfloor = \left\lfloor \frac{n}{2} + \frac{1}{2} \right\rfloor = \frac{n}{2} = \frac{n}{2}$$

qed

$$\sum_{i=0}^n 2^i = 2^{n+1} - 1$$

Proof by induction

$$P(0) \Rightarrow P(1) \Rightarrow \dots \Rightarrow P(n)$$

$$\forall k \in \mathbb{Z}^+ : \left(\sum_{i=0}^{k-1} 2^i = 2^k - 1 \Rightarrow \sum_{i=0}^k 2^i = 2^{k+1} - 1 \right)$$

Proof (5/1/4)

By induction on n .

$$\text{inductive hypothesis (k=1)}$$

$$2^0 + 2^1 + \dots + 2^{k-1} = 2^k - 1$$

Basic step ($n=0$)

$$P(0) \quad 2^0 = 1 = 2^1 - 1$$

inductive step ($k=1$)

$$P(k+1) \quad 2^0 + 2^1 + \dots + 2^{k-1} = 2^k - 1 +$$

$$2^k + 2^1 + \dots + 2^{k-1} + 2^k = 2^k - 1 + 2^k$$

$$\Downarrow \quad 2^0 + 2^1 + \dots + 2^{k-1} + 2^k = 2 \cdot 2^k - 1$$

$$P(k) \quad 2^0 + 2^1 + \dots + 2^{k-1} + 2^k = 2^{k+1} - 1$$

$$\forall n \in \mathbb{Z}^+, n \geq 4 : 2^n < n!$$

inductive hypothesis ($k \geq 5$)

$$2^{k-1} < (k-1)!$$

inductive step ($k \geq 5$)

$$2^{k-1} < (k-1)!$$

$$\Rightarrow 2 \cdot 2^{k-1} < k \cdot (k-1)!$$

$$\Leftrightarrow 2^k < k!$$

$$2^4 = 16 > 4! = 24$$

$$32 > 120$$

$$64 > 720$$